

Clinical Risk Assessment Report: Patient 15

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 13.4% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.60x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which reduced the patient's absolute risk by 2.4%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

- Patient Age (Age)

Value: 45.0 [Bottom 4%] (+2.7%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.
- Small Zone Emphasis (PET) (GLZLM_SZE.PET)

Value: 0.14 [Avg (55th %ile)] (+1.0%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

3. Protective / Mitigating Factors (Reducing Risk)

- Large Zone Emphasis (CT) (GLZLM_LGZE.CT)

Value: 0.02 [Top 24%] (-2.4%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.
- Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)

Value: -0.61 [Bottom 22%] (-1.3%)

Points to continuous, elongated bands of high-density or highly active tumor tissue.
- Carcinoembryonic Antigen (CEA)

Value: 2.1 [Avg (29th %ile)] (-1.0%)

Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.
- Squamous Cell Carcinoma Antigen (SCC_Ag)

Value: 1.5 [Top 6%] (-1.0%)

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.
- 75th Percentile Tumor Density (CONVENTIONAL_HUQ3)

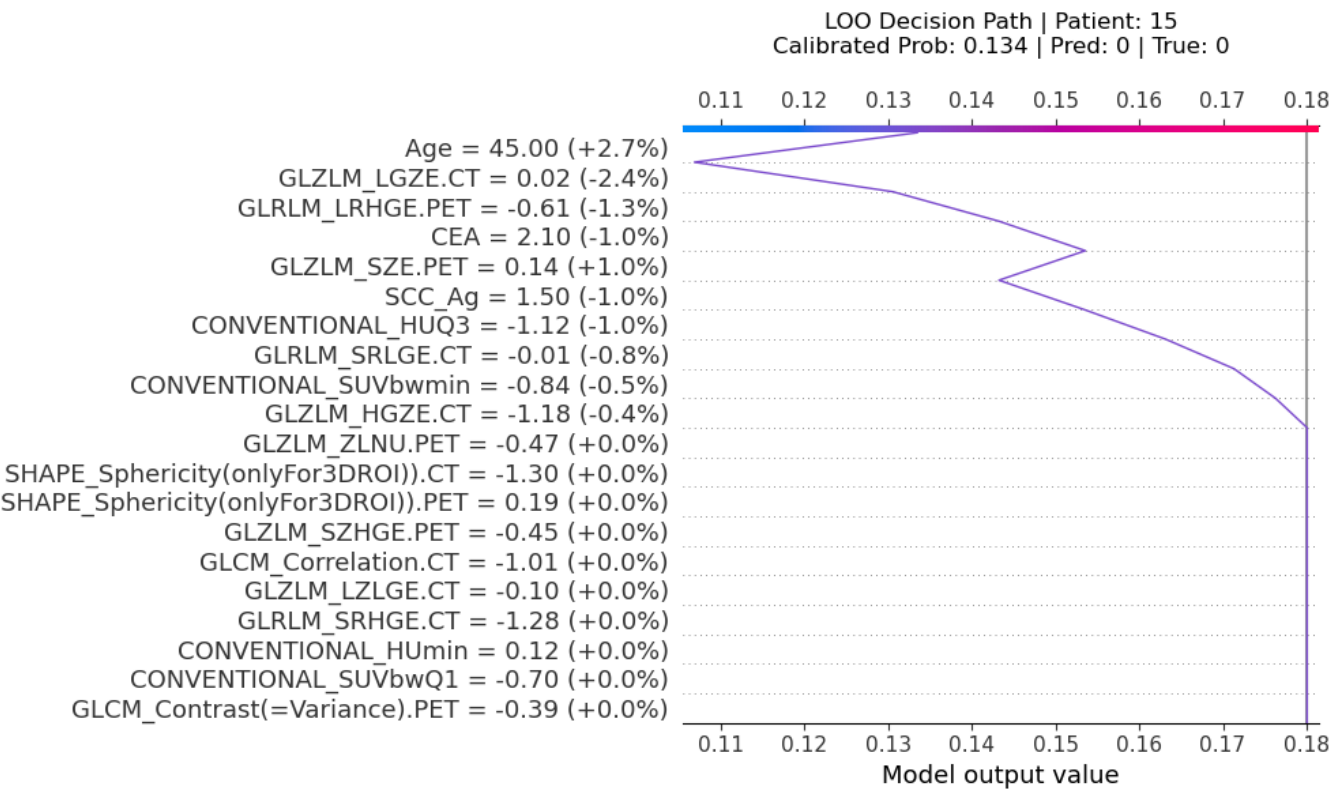
Value: -1.12 [Bottom 14%] (-1.0%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.
- Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)

Value: -0.01 [Top 21%] (-0.8%)

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

